

The BLE node is a low-power peripheral device equipped with an nRF52832 microcontroller, SHT40 temperature/humidity sensor, LIS3DH motion sensor and W25Q16 external flash memory. It can establish a Bluetooth Low Energy connection and transmit sensor data to a gateway or another BLE node.

The nRF52832 is a 64 MHz Arm Cortex-M4F multiprotocol SoC designed for Bluetooth LE and 2.4 GHz applications, with up to 512 KB Flash, 64 KB RAM, NFC-A support, rich digital/analog peripherals and ultra-low-power operation.

Key Features

- Bluetooth Low Energy support with 1 Mbps and 2 Mbps PHY
- 64 MHz Arm Cortex-M4F processor with hardware floating-point unit
- Up to 512 KB Flash and 64 KB RAM for embedded IoT applications
- Integrated NFC-A tag for simplified pairing and tap-to-connect use cases
- Rich peripherals including 12-bit ADC, UART, SPI, TWI (I²C), I2S, PDM, PWM and timers
- EasyDMA support for efficient peripheral-to-memory data movement
- Low-power architecture suitable for battery-powered sensing nodes
- Well suited to wearables, healthcare sensing, beacons, smart-home and environmental monitoring



Fig: BLE Node

Technical Specifications

Parameter	Value
Transceiver	2.4 GHz transceiver with Bluetooth Low Energy support and up to 2 Mbps data rate
CPU	64 MHz Arm Cortex-M4F with FPU and DSP extensions
Memory	Up to 512 KB Flash and 64 KB RAM
Max RF Output Power	Up to +4 dBm
Wireless Range	Up to 100 m (Line-of-Sight, application and antenna dependent)
NFC	Integrated NFC-A tag interface
Supply Voltage	1.7 V to 3.6 V
Operating Temperature	−40 °C to +85 °C
Peripherals	12-bit ADC, UART, SPI, TWI (I ² C), I2S, PDM, PWM, timers and GPIO
DMA	EasyDMA controller for high-efficiency peripheral transfers

Applications

- **Internet of Things (IoT)** — low-power wireless sensor nodes, environmental monitoring, data loggers and smart-home devices.
- **Health and fitness** — wearable activity monitors and connected health-sensing devices.
- **Beacons and remote controls** — location-aware services, BLE beacons and wireless control devices.

- **Smart buildings** — lighting, HVAC, occupancy and security sensing applications.
- **Remote healthcare telemetry** — transmission of sensor measurements such as activity or physiological data to a gateway.